A structural model of emotional dysregulation and Dialectical Behavior Therapy

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Abstract

We embed Dialectical Behaviour Therapy (DBT) inside a continuous-time dynamic-programming model of emotion regulation. An agent who faces stochastic affect shocks divides a finite regulation budget between ineffective coping, which relieves distress immediately but worsens the affect process and accumulates into a stock that lowers income, and skillful coping, which improves the dynamics of affect at the cost of the forgone immediate relief. DBT is the relaxation of a constraint that limits the capacity of skillful coping. We estimate that access to skills is worth about \$16,300 per patient per year in consumption-equivalent terms, with a 90% interval of \$9,200 to \$27,100 that reflects how precisely the treatment effect is identified. The gain is largest for the most severely affected patients. Calibrating the exit cost so that the untreated lifetime suicide probability matches the meta-analytic rate, the model implies that access to skills lowers that probability from about five percent to near zero.

Keywords: Emotional dysregulation, Dialectical Behaviour Therapy, welfare gain, Borderline Personality Disorder

1. Introduction

Dialectical Behaviour Therapy (DBT) is the treatment with established efficacy for borderline personality disorder (BPD), yet after forty years of clinical evidence there is no structural economic model of why it works or how to allocate it. This paper provides one. An agent faces stochastic affect shocks and holds a finite capacity to regulate them. That capacity can go to ineffective coping, which eases distress now but worsens the affect dynamics and builds a stock of dysfunction that lowers income, or, once the agent has learned DBT skills, to skillful coping, which improves the affect process and gives up the immediate relief. DBT is the option to make that substitution. We estimate the primitives from open data and measure the welfare gain from access to skills.

The model nests known structures in the economics of behaviour. The trade-off between immediate relief from ineffective coping and accumulated long-run harm is the rational-addiction dynamic of Becker and Murphy (1988), with ineffective coping in the role of the addictive good and its accumulated effect in the role of the consumption stock. The agent also has an exit option, the economic suicide problem of Hamermesh and Soss (1974) and Becker and Posner (2004). The income channel connects to the applied work on the labour-market consequences of mental illness (Cutler et al., 2006). Our contribution is to place the DBT distinction between ineffective and skillful coping inside this structure and to evaluate skill acquisition.

The model delivers two results. The first is the welfare gain from skills, which the data identify cleanly. The second is the effect of skills on the suicide margin: once the exit cost is calibrated so the untreated lifetime suicide probability matches the meta-analytic figure, access to skills lowers it from about five percent to near zero.

The paper proceeds as follows. Section 2 presents the model; section 3, the estimation procedure; section 4, the results; section 5, a discussion of the results, and section 6 concludes.

2. The Model

2.1. Background

Let c_t be goods consumption, W_t a Brownian motion of emotional shocks, and x_t the endogenous emotional state, with higher x_t better. The agent regulates affect through ineffective coping $z_t \geq 0$ and skillful coping $\xi_t \geq 0$ (Linehan, 1993). The two compete for one finite per-period capacity,

$$z_t + \xi_t \leq B, \qquad z_t, \xi_t \geq 0, \tag{1}$$

so the cost of skillful coping is the immediate relief from the ineffective coping it displaces. Ineffective coping (non-suicidal self-injury, rumination, avoidance, substance use) relieves acute distress but is maladaptive (Linehan et al., 1991; Klonsky, 2007; Nolen-Hoeksema, 1991; Chapman et al., 2006). Skillful coping (mindfulness, distress tolerance, emotion regulation, interpersonal effectiveness) gives no immediate relief and improves the dynamics of affect (Linehan, 1993; Neacsiu et al., 2010). Let Z_t be the stock of past ineffective coping,

$$dZ_t = (\kappa\, z_t - \rho\, Z_t)\, dt, \tag{2}$$

where $\rho \geq 0$ is the decay rate and $\kappa \in (0, 1]$ scales how much momentary coping accumulates into lasting harm, separating the intensity of relief from its persistence. Emotional state evolves as

$$dx_t = \phi(Z_t, \xi_t)[\mu(Z_t, z_t, \xi_t) - x_t]\, dt + \sigma(Z_t, \xi_t)\, dW_t. \tag{3}$$

The attractor of affect depends on coping through the linear form

$$\mu(Z,z,\xi) = \mu_0 + \alpha_\mu(1-\exp(-\xi)) - \beta_\mu Z - \delta z, \tag{4}$$

so skillful coping raises the attractor with saturating returns, the dysfunction stock lowers it at rate β_μ, and ineffective coping lowers it directly at rate $\delta \geq 0$. The last term is what makes such coping ineffective: it buys relief now and worsens the affect process. The speed and volatility are

$$\phi(Z,\xi) = \phi_0 + \alpha_\phi(1-\exp(-\xi)) - \beta_\phi Z, \tag{5}$$
$$\sigma(Z,\xi) = \sigma_0 \exp(\beta_\sigma Z - \alpha_\sigma \xi), \tag{6}$$

with $\phi_0, \sigma_0 > 0$ and all $\alpha_\bullet, \beta_\bullet \geq 0$. Let $T \leq \bar{T}$ be the endogenous exit time. Lifetime utility at t is

$$\mathbb{E}_t \int_t^T \exp[-r(s-t)]\, u(c_s, x_s, z_s)\, ds + \exp[-r(T-t)]\, \omega\, \mathbf{1}_{T<\bar{T}}, \tag{7}$$

with $\omega < 0$ the disutility of exit and $r > 0$ the discount rate. Immediate relief from ineffective coping is bounded,

$$u(c,x,z) = \log c + \gamma \log(1 + x + \lambda(1-\exp(-z))), \tag{8}$$

so coping buys at most λ units of affect-equivalent relief and cannot rescue an arbitrarily bad state; γ weights the emotional channel against consumption. The agent has no credit market and consumes $c_t = \bar{c} - \chi Z_t$ with $\bar{c} > 0$, $\chi \geq 0$, so income falls in the accumulated stock of dysfunction, representing unfavourable academic and labour market outcomes due to the maladaptative behaviours (Skodol et al., 2002; Sansone and Sansone, 2012).

2.2. The constrained problem

Without DBT the agent is constrained to $\xi_t = 0$ and chooses ineffective coping subject to the budget, $0 \leq z \leq B$. Writing V^c for the value function, the HJB in the continuation region is

$$rV^c = \sup_{0\leq z\leq B} \Big\{ u(\bar{c} - \chi Z, x, z) + (\kappa z - \rho Z)\, \partial_Z V^c + \phi(Z,0)[\mu(Z,z,0) - x]\, \partial_x V^c + \tfrac{1}{2}\sigma^2(Z,0)\, \partial_{xx} V^c \Big\}, \tag{9}$$

with the exit option imposed as the lower bound $V^c \geq \omega$ and value matching and smooth pasting at the boundary. The first-order condition for z is

$$\partial_z u + \kappa\, \partial_Z V^c - \delta\, \phi(Z,0)\, \partial_x V^c = \eta^c, \tag{10}$$

with $\eta^c \geq 0$ the multiplier on $z \leq B$ and complementary slackness $\eta^c(z-B) = 0$, $z \geq 0$. The three terms are the marginal relief, the marginal cost of accumulation through $\partial_Z V^c < 0$, and the marginal cost of pulling down the attractor through $\partial_x V^c > 0$. The budget binds, $z = B$, only when relief is valuable enough to outweigh both costs.

2.3. The unconstrained problem

Once the agent learns skills the constraint $\xi = 0$ is lifted and (z, ξ) are chosen jointly subject to (1),

$$rV = \sup_{z+\xi\leq B} \Big\{ u(\bar{c} - \chi Z, x, z) + (\kappa z - \rho Z)\, \partial_Z V + \phi(Z,\xi)[\mu(Z,z,\xi) - x]\, \partial_x V + \tfrac{1}{2}\sigma^2(Z,\xi)\, \partial_{xx} V \Big\}. \tag{11}$$

With $\eta \geq 0$ the multiplier on the budget, the first-order conditions are

$$\partial_z u + \kappa\, \partial_Z V - \delta\, \phi(Z,\xi)\, \partial_x V = \eta, \tag{12}$$
$$[\phi_\xi(\mu - x) + \phi\, \mu_\xi]\partial_x V + \sigma\, \sigma_\xi\, \partial_{xx} V = \eta, \tag{13}$$

where $\phi_\xi = \alpha_\phi \exp(-\xi)$, $\mu_\xi = \alpha_\mu \exp(-\xi)$, and $\sigma_\xi = -\alpha_\sigma \sigma$, with complementary slackness $\eta(z + \xi - B) = 0$ and $z, \xi \geq 0$. At an interior split the two left-hand sides are equal, so the marginal value of capacity is the same in each use and the shadow cost of skills equals the marginal relief from the displaced ineffective coping. By construction $V \geq V^c$, and the welfare gain from access to skills is $V - V^c \geq 0$.

3. Estimation

3.1. Procedure

The structural vector is

$$\Theta = (\phi_0, \alpha_\phi, \beta_\phi, \mu_0, \alpha_\mu, \beta_\mu, \sigma_0, \alpha_\sigma, \beta_\sigma, \rho, \kappa, \delta, \lambda, B, \chi, \bar{c}, \omega, r, \gamma).$$

Five sources identify the primitives. The open experience-sampling archive of cluster B patients (Wright et al., 2017) (228 subjects, 29,024 momentary ratings) gives the baseline affect dynamics $(\phi_0, \mu_0, \sigma_0)$ from a within-person AR(1) on a daily negative-affect composite. The Fisher et al. (2017) experience-sampling study records momentary ineffective coping (rumination, avoidance, reassurance seeking, procrastination) on the same scale as momentary affect, which identifies three objects in affect-standard-deviation units: the budget B as the 95th percentile of coping intensity, the relief ceiling λ as the within-person gap between typical and best affect, and the coping-harm rate δ as the lagged within-person effect of coping on next-period affect. The treatment-effect coefficients are estimated from the Rowland and Wenzel (2020) mindfulness-intervention sample, the between-group difference in the fitted AR(1) giving $\hat{\alpha}_\phi, \hat{\alpha}_\mu, \hat{\alpha}_\sigma$. Because the welfare gain depends almost entirely on α_μ (Section 3.2), we additionally anchor that coefficient to the meta-analysis of standard DBT in BPD by Azevedo et al. (2024) (16 randomised and effectiveness studies, $N = 347$), whose pooled pre-post effect on the Difficulties in Emotion Regulation Scale is $d+ = 1.42$. Treating this uncontrolled figure as an upper bound and mapping the effect size to the attractor, we set $\alpha_\mu = 0.9$ with SE 0.25. The income penalty χ comes from the MIDUS panel (Brim et al., 1995; Ryff et al., 2017, 2019): a regression of log household income on a Five-Factor-derived BPD-features index (standardised neuroticism minus conscientiousness plus a major-depression indicator) with person fixed effects, which removes time-invariant confounders.

The weight γ is the ratio of the affect and log-income coefficients in a MIDUS Ryff well-being regression.

We calibrate $r = 0.04$, $\bar{c} = 1$ at the MIDUS mean income of \$46,072, $\rho = 0.20$ (half-life 3.5 years, matching the 88% ten-year remission rate in Zanarini et al., 2009), and $\kappa = 0.05$ to keep the dysfunction stock in the range implied by the data. The dysfunction-drag coefficients $\beta_\phi, \beta_\mu, \beta_\sigma$ are calibrated; Section 3.2 shows the welfare gain is insensitive to all of them except β_μ, whose influence is bounded. The exit cost ω is calibrated by indirect inference to 0.25, the value at which the untreated lifetime suicide probability is within the meta-analytic 95% confidence interval of Lak et al. (2025). The welfare gain is invariant to ω (Section 5.5), so this calibration affects only the suicide results; the welfare and robustness computations hold ω at a deep floor where the value function is numerically stable. Initial conditions (x_0, Z_0) are drawn from $\mathcal{N}(-1, 0.8^2) \times \mathcal{N}^+(0.8, 0.7^2)$, the distribution of patients at treatment entry. The affect spread 0.8 is the between-person standard deviation of mean affect in the cluster B sample, and the location sets the cohort about one such standard deviation below the sample mean, since patients who enter DBT are drawn from the symptomatic end of the affect distribution rather than its centre. The dysfunction stock Z_0 is positive and dispersed because patients arrive with an accumulated and heterogeneous history of ineffective coping, above the untreated steady state, as presentation is itself triggered by elevated symptoms. The choice matters little for the headline: Table 3 shows the welfare gain is nearly flat across the state space, between \$15,800 and \$16,800 at the eight initial conditions, so it does not turn on the precise onset distribution. The HJB is solved by implicit policy iteration on an 81×41 grid in (x, Z), choosing the joint (z, ξ) optimum under the budget; the simulator uses Euler-Maruyama at $\Delta t = 0.05$ years over 52 years.

3.2. Which coefficients matter

The treatment-effect and drag coefficients are not equally important, so imprecision in most of them is harmless. Table 2 varies each one across its full plausible range and reports the resulting span of the welfare gain. The gain depends on α_μ, the effect of skills on the affect attractor, and to a lesser degree on β_μ. It is flat in α_ϕ, α_σ, β_ϕ, and β_σ, which move it by a few hundred dollars across their whole range. This is why α_μ is the coefficient we anchor to a meta-analysis and why the wide standard errors on α_ϕ and α_σ in Table 1 do not threaten the result. The one calibrated coefficient with material influence is β_μ, and the gain stays between \$13,800 and \$19,000 across its range.

4. Results

We solve the model at $\hat{\Theta}$ with $\bar{c} = 1$ at \$46,072. We present the estimation results in Table 1.

4.1. Welfare gain from DBT

The welfare gain is the permanent transfer $\Delta\bar{c}$ to baseline income that compensates a constrained agent for the absence of skills, the $\Delta\bar{c}$ solving $V^c(x_0, Z_0; \bar{c} + \Delta\bar{c}) = V(x_0, Z_0; \bar{c})$. This is a transfer to the income parameter $\bar{c}$, not to realised consumption $c_t = \bar{c} - \chi Z_t$, so the agent still loses χZ_t to dysfunction. Averaged over the BPD-onset distribution, access to skills is worth \$16,300 per patient per year, about 35% of $\bar{c}$. Table 3 reports the value functions at eight representative initial conditions. The gain is largest at low x_0 and high Z_0, the most severe presentations, which is the basis for allocating DBT by severity.

The interval around this number reflects how well the primitives are identified. A parametric bootstrap that draws $\chi, \gamma, B, \lambda, \delta, \kappa$ and the α coefficients from their sampling distributions gives a 90% interval of \$9,200 to \$27,100, with α_μ accounting for most of the width. Anchoring α_μ to the Azevedo et al. (2024) meta-analysis rather than to the single mindfulness sample is what brings the interval into this range. On a present-value basis at $r = 0.04$ the central figure is roughly \$410,000 per patient. A comprehensive one-to-two-year DBT programme costs on the order of \$10,000 to \$20,000 (Pasieczny and Connor, 2011), so the gain exceeds the cost by a wide margin even under conservative assumptions about durability, consistent with the published cost-effectiveness range of \$25,000 to \$40,000 per quality-adjusted life year.

4.2. Behaviour behind the gain

The constrained agent spends the budget on ineffective coping. That buys bounded relief in flow utility, drags down the affect attractor through δ, and accumulates into Z, which lowers income through χ. The unconstrained agent reallocates part of the budget to skills, trading immediate relief for a better affect process through the α_μ channel. The gain is therefore driven by the dynamics of affect rather than by the income channel, and Section 5.5 shows it is almost unchanged when χ takes its smaller within-person value. Because the worst initial states gain most, the model recommends allocating skills by severity.

4.3. The effect of DBT on suicide

The exit cost ω is the value of the outside option and may be positive. We calibrate it by indirect inference to the untreated lifetime suicide probability. At $\omega = 0.25$ that probability is 5.0%, inside the meta-analytic 95% confidence interval of Lak et al. (2025) (four to eight percent), and the immediate-exit region covers 27% of the state space. Access to skills removes both. The unconstrained lifetime probability is essentially zero and no state is bad enough to make immediate exit optimal. Figure 1 maps the lifetime probability across initial states. Without DBT it is high in the high-Z_0 region, where accumulated dysfunction is severe, and falls sharply to a safe region at low Z_0; skills collapse the whole surface toward zero. The welfare gain is unchanged at this ω because it is invariant to the exit cost, so the two results are separable. The model supports untreated rates only up to about five percent; above that the exit option becomes an attractor for most of the state space and the rate jumps toward one, so we match the lower part of the clinical range rather than its six percent midpoint.

Table 1: Estimated parameter vector $\hat{\Theta}$. SE in parentheses where the parameter is estimated; values without SE are calibrated or normalisations.

parameter	value	source	identification
ϕ_0	4.00[a]	Wright2017 ESM	AR(1) on daily affect
σ_0	0.60[a]	Wright2017 ESM	innovation SD
μ_0	−0.002 (0.0004)	Wright2017 ESM	centred sample mean
B	3.85 (0.19)	Fisher2017 EMA	95th pct coping intensity
λ	0.75 (0.06)	Fisher2017 EMA	within-person affect headroom
δ	0.08 (0.019)	Fisher2017 EMA	lagged coping on next-day affect
κ	0.05	calibrated	bounds the dysfunction stock
α_μ	0.90 (0.25)	Azevedo et al. (2024)	DBT meta-analysis (DERS)
α_ϕ	0.28 (0.32)	Rowland2020 EMA	between-group AR(1)
α_σ	0.51 (1.43)	Rowland2020 EMA	between-group innovation SD
β_ϕ	0.30	calibrated	dysfunction drag on ϕ
β_μ	0.40	calibrated	drag on μ
β_σ	0.30	calibrated	drag on σ
ρ	0.20	Zanarini et al. (2009)	88% remission at 10 yr
χ	0.024 (0.009)	MIDUS 1–3 panel	person fixed effects
$\bar{c}$	1.00	normalisation	MIDUS mean $46,072
γ	0.586 (0.067)	MIDUS well-being	affect / log-income ratio
ω	0.25	indirect inference	untreated suicide rate (Lak et al., 2025)
r	0.04	calibrated	standard annual discount

[a] Rescaled from the raw daily AR(1) estimates $\hat{\phi}_0 = 282.2$ (SE 12.4), $\hat{\sigma}_0 = 6.11$ (SE 0.30) to the year scale of the simulator while preserving the stationary variance. Section 5.5 reports the consistency check.

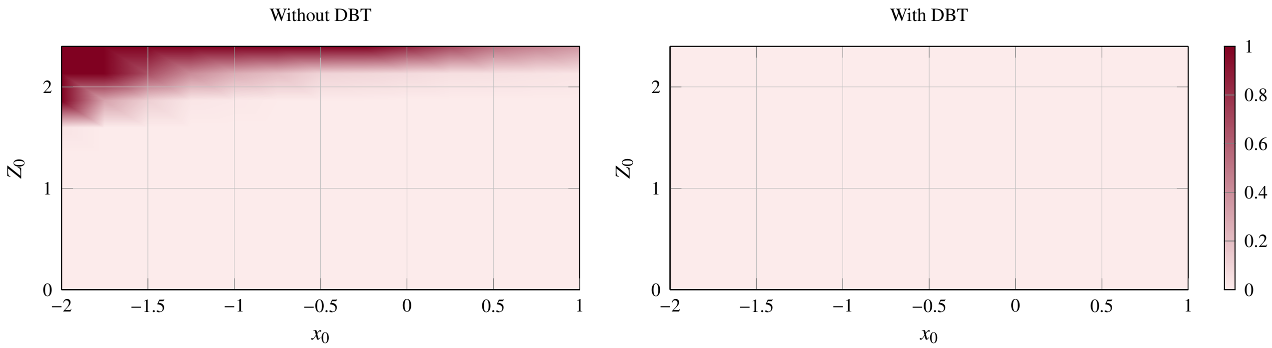

Figure 1: Lifetime suicide probability across initial states (x_0, Z_0) under the constrained (no-DBT) and unconstrained (DBT) policies, at $\omega = 0.25$. Without DBT the probability is concentrated at high Z_0 and falls sharply to a safe region at low Z_0. Access to skills collapses the surface toward zero.

Table 2: Welfare gain as each treatment-effect and drag coefficient varies over its plausible range, other parameters at $\hat{\Theta}$.

coefficient	range	welfare gain ($/yr)
α_μ	[0.4, 1.4]	7,800 to 24,000
β_μ	[0.0, 0.8]	13,700 to 18,900
α_σ	[0.0, 1.0]	16,000 to 16,300
β_σ	[0.0, 0.6]	16,100 to 16,500
α_ϕ	[0.0, 1.0]	16,200 to 16,300
β_ϕ	[0.0, 0.6]	16,200 to 16,300

5. Discussion

5.1. Magnitude of the welfare gain

The central figure of $16,300 per patient per year is the permanent transfer to baseline income that compensates the constrained patient for not having skills. It is not the gain in realised consumption, which is smaller because the agent still loses χZ_t. The estimate is large relative to the cost of DBT, and its order of magnitude matches the published cost-effectiveness literature, so the model places the value of skills in a credible range.

5.2. Identifying the treatment effect

The width of the welfare interval is governed by how well the effect of skills on the affect attractor is identified. A single open mindfulness sample puts almost no discipline on it. The Azevedo et al. (2024) meta-analysis of standard DBT in BPD, pooling sixteen studies, tightens that one coefficient enough to give a 90% welfare interval of $9,200 to $27,100. The remaining treatment-effect and drag coefficients are immaterial to the gain (Table 2), so their wide standard errors do not matter. Two sources of residual uncertainty exist: the meta-analytic effect is a pre-post change in an emotion-regulation scale rather than a direct estimate of the attractor, and the mapping from that scale to the model parameter is a judgement. The interval reflects both.

Table 3: Welfare gain from DBT skills by initial condition, at $\hat{\gamma}$.

x_0	Z_0	V^c	V	$V - V^c$	\$/year
−1.5	0.5	3.11	10.64	7.53	16,194
−1.0	0.5	3.24	10.72	7.48	16,079
−0.5	0.5	3.31	10.78	7.46	16,022
−1.5	1.0	2.55	10.33	7.78	16,823
−1.0	1.0	2.73	10.42	7.69	16,604
−0.5	1.0	2.82	10.48	7.66	16,521
−1.0	0.2	3.51	10.90	7.38	15,824
0.0	0.5	3.38	10.82	7.44	15,980

Dollar figures are permanent transfers to baseline income $\bar{c}$ that equalise the constrained and unconstrained value functions. The conversion $\Delta\bar{c}/\bar{c} = \exp(r(V - V^c)) - 1$ holds under separable utility because $\log c$ enters with unit coefficient.

5.3. The exit option and the suicide margin

The exit cost ω is the value of the outside option and need not be negative, so we calibrate it by indirect inference to the untreated suicide probability. The model supports rates only up to about five percent. Above that the exit option becomes an attractor for most of the state space and the rate jumps toward one, so we match five percent, the lower part of the meta-analytic range, rather than its six percent midpoint. We are cautious about the level, since a forward-looking optimum is an imperfect model of impulsive acts, but the comparative result is robust: access to skills removes essentially all of the suicide risk and the entire immediate-exit region (Figure 1). Because the welfare gain is invariant to ω, the welfare and the suicide results are separable, and we compute the welfare objects at a deep floor where the value function is numerically stable.

5.4. A rational model of a behavioural constraint

The agent is rational and discounts exponentially, so DBT here relaxes a constraint rather than correcting a bias. A present-biased version along the lines of Gruber and Kőszegi (2001) would give skills an extra internality-correcting value, since a hyperbolic agent over-uses immediate relief. We leave that extension to later work and read our estimate as a lower bound on the value of skills for that reason. The no-credit-market assumption is the standard hand-to-mouth specification (Cutler et al., 2006) and is supported for BPD by high rates of disability income and limited labour-market access (Zanarini et al., 2009).

5.5. Robustness

The income penalty illustrates why the result rides on the affect channel. The cross-sectional association of the BPD-features index with income overstates the causal penalty: the person-fixed-effects estimate is $\chi = 0.024$ (SE 0.009) and the first-difference estimate is 0.012, against a pooled value near 0.08. The welfare gain moves by under 4% across this whole range. The headline is similarly insensitive to the discount rate (\$16,213 at $r = 0.03$ to \$16,257 at $r = 0.05$) and to the persistence of dysfunction (\$18,123 at $\rho = 0.10$ to \$15,419 at $\rho = 0.30$), and it is invariant to the exit floor ω across $[-25, 0]$. The time-scale rescaling is a change of units: re-solving at the raw daily estimates $\hat{\phi}_0 \approx 282$, $\hat{\sigma}_0 \approx 6.11$ on a finer grid gives a gain within 6% of the rescaled baseline, and a variance-matched rescaling within 3%. Sensitivity to the emotional weight γ is reported in Table 4; the estimated $\hat{\gamma} \approx 0.59$ sits between the low and high calibrations.

Table 4: Welfare gain under different γ, at $(x_0, Z_0) = (-1, 0.5)$.

spec	γ	$V - V^c$	pct	\$/year
additive	n/a	12.80	67%	30,814
separable	1.00	12.75	66%	30,637
separable (estimated)	**0.59**	**7.48**	**35%**	**16,079**
separable	0.50	6.39	29%	13,420
separable	0.25	3.21	14%	6,315

6. Conclusion

We have written a structural model of DBT as the relaxation of a regulation budget, estimated its primitives from open experience-sampling data, a DBT meta-analysis, and the MIDUS panel, and measured the welfare gain from access to skills. The gain is about \$16,300 per patient per year in consumption-equivalent terms, with a 90% interval of \$9,200 to \$27,100, and it is largest for the most severely affected patients. Calibrated to the untreated suicide probability, the model also implies that access to skills lowers the lifetime suicide rate from about five percent to near zero. The marginal dollar of mental-health spending is best deployed on DBT access for those patients.

Appendix A. Baseline-income equivalent welfare conversion

The welfare gain $V - V^c$ is in utility units. We convert it by asking what permanent transfer $\Delta\bar{c}$ to the income parameter $\bar{c}$ compensates the constrained agent. Under (8) consumption enters only the first term, so by the envelope theorem

$$\partial V/\partial \bar{c} = \mathbb{E}_0 \int_0^T \exp(-rs)(\bar{c} - \chi Z_s)^{-1} ds. \tag{A.1}$$

Replacing $1/(\bar{c} - \chi Z_s)$ by $1/\bar{c}$ gives $V^c(\bar{c} + \Delta\bar{c}) - V^c(\bar{c}) \approx r^{-1}\log((\bar{c} + \Delta\bar{c})/\bar{c})$, and equating to $V - V^c$,

$$\frac{\Delta\bar{c}}{\bar{c}} = \exp[r(V - V^c)] - 1. \tag{A.2}$$

The approximation upper-bounds the true transfer, since $1/(\bar{c} - \chi Z_s) \geq 1/\bar{c}$ when $Z_s > 0$; at our parameters the gap is a few percent of the reported figure.

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